

Analyzing Non-Performing Loans in Finland | Cloud Data Analytics

Prepared for: Portfolio Presentation

Overview

This project focuses on analyzing the Finnish loan portfolio with a particular emphasis on non-performing loans (NPL). The goal is to provide clear insights into loan quality, sector-level risk, and changes in NPL ratios over time. A report and interactive dashboard were developed to visualize key indicators such as total loans, NPL levels, sector risk distribution, and overall credit-risk trends.

Objective

- Target Goal #1:** Conduct an analysis of the NPL dataset to understand credit risk across major economic sectors from 2014–2018. The analysis focuses on total loan exposure, non-performing loan amounts, and NPL ratios to identify high-risk areas.
- Target Goal #2:** Build an interactive dashboard enabling the risk and treasury teams to explore sector-level credit performance, compare risk levels, and monitor trends over time.
- Impact:** This project provides actionable insights to support risk mitigation, capital allocation, and early-warning monitoring across the loan portfolio.

Next Steps

- ➔ **Collect feedback from users about how easy the dashboard is to understand and use.** This feedback can help identify what should be improved, such as clearer charts, better labels, or more intuitive navigation.
- ➔ **Add more detailed analysis of sectors or borrower groups to deepen the insights.** This additional segmentation can help highlight where the highest risks are and support more informed decisions about credit management.

Results

- The results of the data analysis demonstrate that the NPL ratio remained broadly stable between 2014 and 2018, with only minor year-to-year changes.

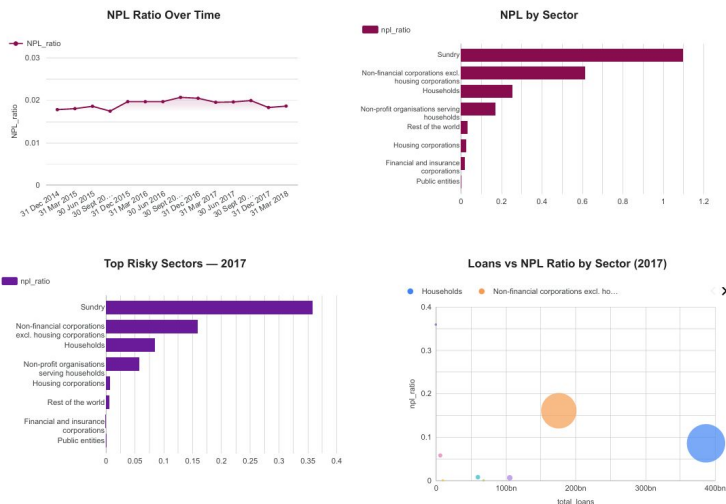
Figure 1 – NPL Ratio by Year (2014–2018)

Year	total_loans	total_npl	npl_ratio
2015	920,526,891,132	16,887,470,068	0.54
2016	867,036,874,770	17,386,174,527	0.86
2017	813,935,403,848	15,682,570,336	0.68
2014	228,701,146,433	4,052,779,032	0.07
2018	204,027,739,910	3,787,760,285	0.09

Note: Table shows the total loan exposure, total non-performing loans, and NPL ratio for each year from 2014–2018. The highest loan exposure was €920.5B in 2015, and the lowest was €204.0B in 2018.

- Data analysis highlights three core aspects of the Finnish loan portfolio—total loan exposure, non-performing loan levels, and sector-specific risk patterns—summarized on the interactive dashboard.

Figure 2 – Interactive Dashboard



Note: The dashboard includes a time-series line chart, sector-level bar charts, and a loans-versus-NPL scatter visualization.